

# Expert Use Guide

Operational semantics, completion predicates and evidence flow

## What this workbench does

It turns a mission into a connected early engineering model: stakeholder needs and objectives lead to requirements; functions and components describe the solution; parameters, verification and simulations provide evidence; configurable alternatives can be compared and approved. It supports preliminary programme and architecture work, not certified detailed-design analysis.

### Two synchronized ways to work

**Architect view** One guided question at a time, with answer guidance, examples, progress and readiness.

**Modeler view** Dashboard plus direct tables, graphs, matrices, formulas, variability, analysis and delivery.

**Both views update the same canonical project; switching never creates a second model.**

## Choose a scope

|                             |                                                                |
|-----------------------------|----------------------------------------------------------------|
| Architecture definition     | Traceable product and industrial-system architecture           |
| Architecture and simulation | Architecture plus KPI evidence for the current model           |
| Trade-off                   | Configurable 100% alternatives, comparable runs and a decision |

For practitioners with MBSE / MBPLE knowledge.

Current branch: 03\_Architect\_view\_v01 - application v1.5.3 - schema 9

Guide refreshed 4 September 2026. The introduction occupies this single page.

**Continue with the existing guide, then use the appended answer handbook for every Architect question group.**

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## Architect view - guided model creation

The Architect view is an alternate authoring interface over the same canonical project. Answers create or link model elements, directed relationships, parameters, formulas, configurations, runs and decisions; the Modeler view remains the detailed editor.

### Choose a perspective

| Perspective    | Use it for                                                                    | Effect                                                             |
|----------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Architect view | Guided creation and recap through one question at a time.                     | Confirmed answers write canonical project content.                 |
| Modeler view   | Direct editing of tables, matrices, graphs, formulas and analysis workspaces. | Edits use the same project and can mark guided answers for review. |

### Work through the questions

|          |                                                                                                                                                                                                  |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1</b> | <b>Select the analysis scope</b><br><br>Choose Architecture definition, Architecture and simulation, or Trade-off study. The choice controls which later questions appear.                       |
| <b>2</b> | <b>Answer one question</b><br><br>Type or select the requested information, then click Continue. The answer appears in the progress panel and creates or links the corresponding model content.  |
| <b>3</b> | <b>Use Skip only for a visible gap</b><br><br>Skip has no timer and never advances automatically. A skipped required answer remains visible and prevents the study from being reported as Ready. |
| <b>4</b> | <b>Return and revise safely</b><br><br>Use Back or the progress panel to reopen an answer. If a change affects generated content, review the impact message before applying it.                  |
| <b>5</b> | <b>Resume later</b><br><br>The current question and confirmed answers are stored with the project. Save Project for a portable JSON backup; loading the project resumes the guided session.      |

## Question order shared by all scopes

| Stage                      | What you provide                                                                                             | What is created                                                                  |
|----------------------------|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Study setup                | Scope, project name and mission.                                                                             | Project context and the main purpose of the study.                               |
| Stakeholder intent         | All stakeholders, then the system of interest, needs and objectives.                                         | Mission and stakeholder traceability.                                            |
| Use cases and requirements | Use cases; quantitative requirements; parameter, operator, target and unit; engineering values where needed. | Need/objective -> requirement traces, formulas, bindings and satisfaction links. |
| Product architecture       | Functions, components, realizations and relevant satisfaction evidence.                                      | Use-case behavior and product technical architecture.                            |
| Industrial architecture    | Process functions, equipment, product flows, durations and resources.                                        | Industrial behavior, sequence, components and resource evidence.                 |
| Review                     | Confirm traceability, requirements and architecture recaps.                                                  | A concise completeness record and visible unresolved gaps.                       |

## Questions added by scope

| Scope                       | Additional guided stages                                                                                                                                 | Final recap                                                        |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Architecture definition     | None beyond the shared architecture path.                                                                                                                | Overview, stakeholder traceability, architecture and requirements. |
| Architecture and simulation | KPI selection/formulas, missing engineering inputs and simulation.                                                                                       | Shared recaps plus current simulation evidence.                    |
| Trade-off study             | Decision question immediately after requirements; common/alternative characteristics; variability, configurations, simulations, comparison and decision. | Shared and simulation recaps plus comparison and baseline.         |

### Requirement satisfaction rule

Every requirement must be satisfied by at least one function, technical component or parameter. Individual functions and components do not each need to satisfy a requirement.

## Recap and synchronization

The final pages summarize the study purpose, stakeholder-to-evidence chain, architecture, requirement state and - when relevant - simulation and trade-off evidence. Detailed formulas, matrices and element records remain available in Modeler view.

### Review after direct editing

If the model is changed in Modeler view, affected guided answers are marked for review instead of silently remaining current.

# Operating model

The workbench is a browser-based preliminary engineering environment. Its value is the explicit evidence chain from intent and scope through product and industrial architectures, variant derivation, simulation and trade decision. The dashboard workflow is the authoritative route map.

## Authority boundary

Treat results as preliminary engineering estimates. The workbench supports traceability, reproducible calculations and decision evidence; it is not a detailed CAD, production planning or certified verification environment.

## Workspace responsibilities

| Workspace           | Authoritative content                                                                         | Downstream consumers                                                       |
|---------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Model               | Elements, properties, requirements and canonical directed relationships.                      | Variability mappings, parameter bindings, simulations, traceability views. |
| Variability         | Feature model, constraints, variation points, configurations and immutable 100% realizations. | Configured simulation and trade eligibility.                               |
| Parameters and KPIs | Typed parameters, process duration, resource quantities and KPI definitions.                  | Requirement evaluation and simulation.                                     |
| Simulation          | Immutable input snapshots, KPI results, warnings, assumptions and breakdown.                  | Trade feasibility and ranking.                                             |
| Trade Study         | Study scope, common KPI basis, current comparison and explicit decision.                      | Approved baseline evidence.                                                |
| Export              | Portable project JSON and selected data/report packages.                                      | Review, transfer and audit.                                                |

## Canonical trace semantics

| Source                      | Relation                          | Target                              |
|-----------------------------|-----------------------------------|-------------------------------------|
| Mission                     | hasStakeholder                    | Stakeholder                         |
| Stakeholder                 | hasNeed / hasObjective            | Need / Objective                    |
| Stakeholder                 | involvedIn                        | Use Case                            |
| Use Case                    | hasFunction                       | Product Function / Process Function |
| Need / Objective            | derives                           | System Requirement                  |
| Requirement                 | satisfiedBy                       | Function / Component                |
| Verification Method         | verifies                          | Requirement                         |
| Product Function            | realizedBy                        | Product Component                   |
| Process Function            | realizedBy                        | Industrial System Component         |
| Process Function            | allocatedTo / consumes / produces | Product Component                   |
| Industrial System Component | requiresResource                  | Resource                            |

## Resource allocation rule

requiresResource terminates at the industrial system component that realizes the process function. Resource assignments are not placed on the function. Required quantity belongs to the relation.

# Project lifecycle and evidence invalidation

The project header controls project identity, architecture context, revision and persistence. Browser-local autosave is convenient state, while Save Project is the portable complete artifact.

| Change                               | What becomes non-current                                                | Required action                                                                        |
|--------------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Model element, relation or parameter | Affected requirement evidence, 100% realizations, runs and comparisons. | Revalidate formulas, rederive affected configurations, rerun and recompute comparison. |
| Feature model or variation point     | Affected configurations / realizations and their downstream runs.       | Validate configurations and rederive.                                                  |
| KPI definition or selected study KPI | Runs and trade comparisons using the old definition or basis.           | Rerun every alternative with the common current KPI set.                               |
| Study scope, weights or objective    | Existing comparison.                                                    | Retain old immutable result; create a new current comparison.                          |

The screenshot displays the MBSE / MBPLE Programme workbench interface. On the left is a dark sidebar with a navigation menu. The main area shows project controls at the top, including 'ACTIVE PROJECT' (Configurable Coffee Machine Product Line), a dropdown for 'Configurable Coffee Machine Product L', and buttons for '+ New project', 'Duplicate', and a trash icon. Below these are 'Save Project' and 'Load Project' buttons, and a dropdown for '150% Product-Line Model'. A status bar indicates 'Local autosave: Saved', 'Revision 1', and buttons for 'Reset sample' and 'Reset all'.

The main content area is titled 'Snapshots and Delivery' (Stage C) and includes a warning: 'Preliminary engineering estimate — not a verified detailed-design result.' Below this is a section for 'Save / Load Project' with tabs for 'Snapshots', 'Selective Export', 'PDF Report', and 'Demo Checklist'.

The 'Save Project' section describes downloading the complete active project as a portable, validated JSON package. It shows the project name 'Configurable Coffee Machine Product Line' and schema 'Schema 9 · revision 1 · 41 model elements'. A note states: 'The project file includes model, variability, configurations, derivations, simulation evidence, Trade Studies and decisions. Application-level snapshots and interface preferences remain local and are not embedded.' A 'Download project file' button is present.

The 'Load Project' section describes validating a saved project file before loading. It includes a 'Choose File' button and a 'No file chosen' message. Under 'LOAD BEHAVIOR', there are two options: 'Load as new project' (selected) and 'Replace active project'. The 'Load as new project' option has a description: 'Keep the active project and resolve only a top-level project-ID collision.' The 'Replace active project' option has a description: 'Preserve the active project ID and create a safety snapshot first.' A 'Validate and load' button is at the bottom.

Project controls expose local autosave, revision, architecture context and complete JSON save/load.

## Model views as verification instruments

- Editable table: authoritative create and edit workflow.
- Context / trace graph: local topology and directionality.
- Traceability matrix: cross-domain completeness and rapid relation editing.
- Model quality: machine-checkable gaps and structural diagnostics.
- Sequence diagram: ordered behavior for function domains.

Save Project

Load Project

150% Product-Line Model

Local autosave: Saved

Revision 1

Reset sample

Reset all

Build the mission-to-validation thread in any order; the workflow and validation remain live.

Mission

+ Add row

+ Add column

Units

Mission and Context

Requirements and Validation

Product Functional

Product Technical

Process Functional

Process Technical

Interfaces and T...

Search elements, tags, parameters...

All types

All statuses

150% Product-Line Model

All scopes

Filter tag

Editable table

Context graph

Traceability matrix

Model quality

Manual order

13 of 41 elements

| Name                                                                                          | Type        | Working scope | Status   | Authoritati... |
|-----------------------------------------------------------------------------------------------|-------------|---------------|----------|----------------|
| <div>⋮</div> <div>Deliver a configurable coffee experience</div> <div>Open full details</div> | Mission     | —             | reviewed | —              |
| <div>⋮</div> <div>Coffee Machine System</div> <div>Open full details</div>                    | Stakeholder | —             | reviewed | —              |
| <div>⋮</div> <div>Product Manager</div> <div>Open full details</div>                          | Stakeholder | —             | reviewed | —              |
| <div>⋮</div> <div>Coffee Consumer</div> <div>Open full details</div>                          | Stakeholder | —             | reviewed | —              |
| <div>⋮</div> <div>Service and Quality Manager</div> <div>Open full details</div>              | Stakeholder | —             | reviewed | —              |

Use the matrix to expose missing cross-domain traces; use graph and quality views to confirm direction and structure.

# Chapter 1 - Architecture building

Objective: produce a scoped, trace-complete product and industrial system architecture. Simulation, configurations and trade comparison are outside this scope.

## Problem-space execution

| Step                              | Required information                                                                                                            | Completion predicate / impact                                                                                                                       |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Mission, system and intent     | Mission; stakeholders; exactly one system-of-interest flag; needs; objectives; hasStakeholder, hasNeed and hasObjective traces. | At least one mission has a stakeholder; one and only one stakeholder is the SOI; at least one need and objective exist. Establishes intent lineage. |
| 2. Use-case and requirement scope | Use cases; stakeholder involvement; active-scope selection; requirements derived from scoped needs/objectives.                  | Every selected use case involves the SOI; at least one scoped need/objective derives a requirement. Controls all later completeness checks.         |

## Solution-space execution

| Step                                                       | Required information                                                                                                                                        | Completion predicate / impact                                                                                                                               |
|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Product functional and technical architecture           | Product functions per scoped use case; product components; realizedBy allocation; requirement satisfaction where applicable.                                | Every selected use case has product functions and every product function has at least one realizing product component.                                      |
| 2. Industrial system functional and technical architecture | Process functions per scoped use case; industrial system components; process-to-equipment realizedBy; product flow; component-to-resource requiresResource. | Every selected use case has process functions and every process function has a realizing industrial component. Resources are allocated to those components. |

## Recommended execution order

|                                                                                                                                                                                                      |                                                                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1</b>                                                                                                                                                                                             | <b>Create and scope intent</b><br>Author mission, stakeholder/SOI, need, objective, use case and requirement elements. Confirm canonical relation direction in graph view.                          |
| <b>2</b>                                                                                                                                                                                             | <b>Build product behavior before product structure</b><br>Decompose scoped services into functions, then allocate functions to components. Avoid orphan structure with no behavioral justification. |
| <b>3</b>                                                                                                                                                                                             | <b>Build process behavior before industrial structure</b><br>Model process functions, precedence and product flow, then allocate to equipment/workstations.                                         |
| <b>4</b>                                                                                                                                                                                             | <b>Attach resources and quantities</b><br>Create requiresResource from industrial components and store quantity on the assignment relation.                                                         |
| <b>5</b>                                                                                                                                                                                             | <b>Run closure checks</b><br>Use matrix and Model quality to prove intent-to-architecture and use-case-to-resource chains. Resolve or document every gap.                                           |
| <b>6</b>                                                                                                                                                                                             | <b>Freeze a portable checkpoint</b><br>Download complete project JSON and, if required, selective XLSX/JSON or report output.                                                                       |
| <b>Scope isolation</b><br>Dashboard completeness is evaluated against the selected use-case scope. Keep future or unrelated use cases unselected rather than satisfying them with artificial traces. |                                                                                                                                                                                                     |

The screenshot displays the MBSE / MBPLE Programme workbench interface. On the left is a dark sidebar with navigation options: Dashboard, PROBLEM SPACE (2/2), SOLUTION SPACE (2/2), and WORKSPACES (Model, Variability, Parameters and KPIs, Simulation, Architecture Trade Study, Export). The main area shows the 'ACTIVE PROJECT' 'Configurable Coffee Machine Product Line' with a dropdown menu set to 'Configurable Coffee Machine Product L'. Below this are buttons for 'Save Project', 'Load Project', and a '150% Product-Line Model' dropdown. A status bar indicates 'Local autosave: Saved', 'Revision 1', and buttons for 'Reset sample' and 'Reset all'. The 'PROJECT WORKFLOW' section is titled 'Configurable Coffee Machine Product Line' and includes a warning: 'Preliminary engineering estimate — not a verified detailed-design result.' It instructs users to 'Choose the project outcome, then complete the model activities shown in Problem Space and Solution Space.' The 'Overall scope' section shows three options: 'Architecture building' (selected), 'Architecture building + simulation', and 'Trade study'. A status bar below indicates 'Status: Not started, In progress, Complete, Blocked'. The 'Problem Space' section shows two steps: '1 Define mission, system and intent' and '2 Define use-case and requirement scope', with a '2/2 complete' indicator.

The Architecture building dashboard is the master sequence and exposes blocked or incomplete predicates.

## Architecture-building acceptance checklist

| Control                 | Acceptance evidence                                                            |
|-------------------------|--------------------------------------------------------------------------------|
| Intent coverage         | Each scoped requirement has a visible derivation from a need or objective.     |
| Functional coverage     | Each selected use case has both product and process behavior where applicable. |
| Allocation completeness | No scoped product or process function is without a realizing component.        |
| Resource semantics      | No resource is allocated directly to a process function.                       |
| Directionality          | Canonical source, relation and target directions match the relationship table. |
| Quality                 | No unexplained blocker remains in the dashboard or Model quality view.         |

# Chapter 2 - Architecture building + simulation

Objective: establish the architecture from scoped intent, complete quantitative engineering inputs, evaluate requirements and create a current simulation result for the selected KPI set.

## Workflow and completion semantics

| Phase        | Activities                                                               | Result                                                          |
|--------------|--------------------------------------------------------------------------|-----------------------------------------------------------------|
| Problem 1-2  | Mission/SOI/intent, active use cases and derived requirements.           | Same predicates as Architecture building.                       |
| Problem 3    | Define evaluation KPI metadata and formula or standard algorithm.        | At least one evaluation KPI with calculable basis.              |
| Solution 1-2 | Build fully allocated product and industrial architectures.              | Same architecture predicates as Architecture building.          |
| Solution 3   | Complete component parameters, process duration and resource quantities. | All selected KPI dependencies and requirement bindings resolve. |
| Solution 4   | Validate and run architecture-only simulation.                           | Current run with numeric results for all selected KPIs.         |

## Parameter model

Numeric engineering values are typed parameters on product or industrial components. A sound parameter record contains name, semantic key, numeric value, unit, origin, provenance, range or uncertainty and configuration applicability.

| Concern          | Practice                                                                                     | Impact                                                         |
|------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Semantic key     | Use mass, directCost or power when the matching standard algorithm should collect the value. | Controls algorithm discovery; labels alone are not sufficient. |
| Unit             | Use a compatible explicit unit for every numeric value.                                      | Formula validation and KPI dimensional consistency.            |
| Origin           | Distinguish entered, assumed, calculated and simulated.                                      | Makes uncertainty and authority visible.                       |
| Provenance       | Record source, revision and relevant conditions.                                             | Supports review of snapshot inputs.                            |
| Applicability    | No selection means all configurations; otherwise specify variants.                           | Controls active 100% model inputs.                             |
| Calculated value | Use @references from the same component or transitive refiners.                              | Maintains a model-resident calculation graph.                  |

Save Project

Load Project

150% Product-Line Model

Local autosave: Saving

Revision 1

Reset sample

Reset all

## Parameters and KPIs

Editable engineering inputs, exact applicability, safe formulas, units, and standard pre-sizing algorithms.

Engineering inputs

KPI definitions

Engineering input table

Blank applicability means every configuration. Value-origin classification and source provenance are independently recorded.

Search owner, name, :

+ Parameter

| Owner / parameter                                                                                          | Semantic key | Value      | Range            | Origin                                        | Applicability                                |                             |
|------------------------------------------------------------------------------------------------------------|--------------|------------|------------------|-----------------------------------------------|----------------------------------------------|-----------------------------|
| <b>Throughput</b><br>Ingredient-handling module · par-productcomponent-material-handling-module-throughput | throughput   | 12 parts/h | — ... —          | assumed<br>Programme preparation assumption   | Every configuration                          | <div>Edit</div> <div></div> |
| <b>Mass</b><br>Brewing and heating module · par-productcomponent-joining-module-mass                       | mass         | 420 kg     | 300 ... 700      | assumed<br>Programme preparation assumption   | configuration-manual, configuration-balanced | <div>Edit</div> <div></div> |
| <b>Automated mass</b><br>Brewing and heating module · par-automated-mass                                   | mass         | 650 kg     | 500 ... 800      | entered<br>Robot-cell supplier budgetary data | configuration-automated                      | <div>Edit</div> <div></div> |
| <b>Manual direct cost</b><br>Brewing and heating module · par-manual-cost                                  | cost         | 35000 EUR  | 25000 ... 50000  | assumed<br>Programme preparation assumption   | configuration-manual, configuration-balanced | <div>Edit</div> <div></div> |
| <b>Automated direct cost</b><br>Brewing and heating module · par-automated-cost                            | cost         | 90000 EUR  | 70000 ... 120000 | entered<br>Robot-cell supplier budgetary data | configuration-automated                      | <div>Edit</div> <div></div> |

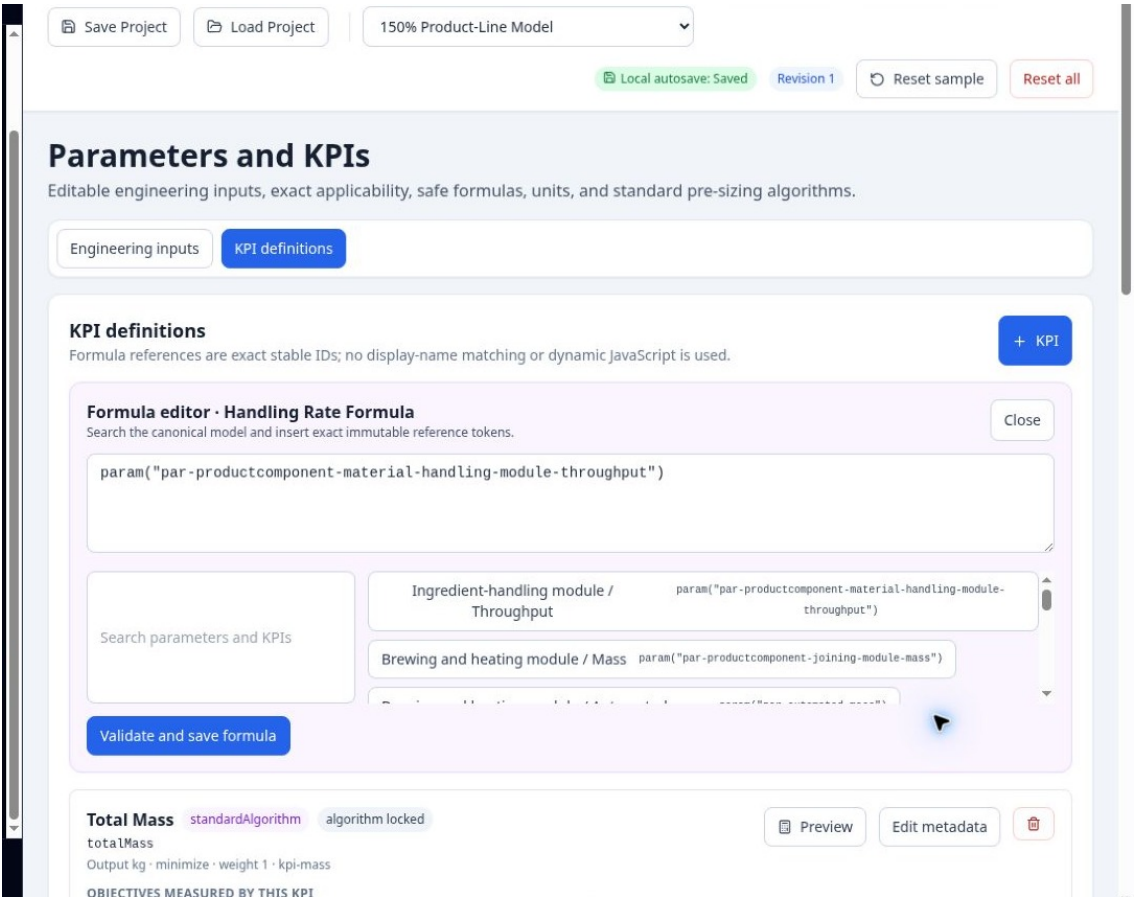
Process duration and component-to-resource quantities are authoritative inputs for time, cost and utilization algorithms.

## Formula layers and resolution

| Layer                | Syntax                                                     | Resolution and constraints                                                                                                                    |
|----------------------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Calculated parameter | @symbol; + - * /; parentheses; sqrt; ^ or **.              | Same component and transitive refiners. Optional requested output unit.                                                                       |
| Requirement          | Arithmetic plus exactly one comparison: < <= > >= = == !=. | @symbol bindings store immutable source IDs. Missing or pending evidence never silently passes.                                               |
| KPI                  | param("id"), kpi("id"), arithmetic, min/max/sum/average.   | Stable-ID references across active model. Addition and aggregation require exact unit equality. KPI dependencies must be selected in the run. |

### Parameter collection

The KPI builder searches numeric parameters by name, unit, owning element, element type and tags, then inserts the immutable parameter ID. The simulation resolves only parameters active in the selected architecture or configuration. Arbitrary table columns are not calculation inputs.



The builder creates stable-ID KPI expressions and shows the model source behind each token.

KPI definition protocol

|   |                                                                                                                                                                              |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>Define decision meaning</b><br>Set name, objective measured, direction (minimize/maximize), nonnegative global weight and output unit.                                    |
| 2 | <b>Choose calculation authority</b><br>Use a standard algorithm only when its required semantics match. Otherwise create a formula from model parameters and dependent KPIs. |
| 3 | <b>Validate units and dependencies</b><br>Ensure unit-compatible operations and select every dependent KPI in the run.                                                       |
| 4 | <b>Review provenance</b><br>Use the run breakdown to confirm that only intended active parameters were collected.                                                            |

## Standard algorithm inputs

| Algorithm             | Collection / calculation rule                                                                                                                                                  |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| totalMass             | Sum active numeric parameters whose semantic key is mass.                                                                                                                      |
| directElementCost     | Sum active numeric parameters whose semantic key is directCost.                                                                                                                |
| processCost           | For each industrial component resource assignment: durations of realized process functions x resource hourlyRate x requiredQuantity. Missing quantity warns and defaults to 1. |
| estimatedTotalCost    | directElementCost + processCost.                                                                                                                                               |
| totalPower            | Sum active numeric parameters whose semantic key is power.                                                                                                                     |
| manufacturingLeadTime | Critical longest path over processFunction precedes relations; 1 day = 8 hours.                                                                                                |
| resourceDemand        | Duration x requiredQuantity per industrial component resource assignment, with breakdown.                                                                                      |
| basicUtilization      | Resource demand / (capacity hours x availability percent).                                                                                                                     |
| throughputProxy       | 1 / manufacturingLeadTimeHours; comparative proxy only.                                                                                                                        |

## Simulation execution protocol

|          |                                                                                                                                                                    |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1</b> | <b>Select architecture-only 150% mode</b><br><br>For this scope, choose the current architecture context, name the run and select the complete KPI dependency set. |
| <b>2</b> | <b>Validate inputs</b><br><br>Resolve missing bindings and unit errors. Review warnings and explicitly acknowledge only acceptable assumptions.                    |
| <b>3</b> | <b>Execute and inspect snapshot</b><br><br>Verify numeric result, unit, source values, assumptions, warnings, breakdown and critical chain where applicable.       |
| <b>4</b> | <b>Confirm currency</b><br><br>Any relevant input change makes the run stale. Rerun before using results as evidence.                                              |

Save Project

Load Project

150% Product-Line Model

Local autosave: Saved

Revision 1

Reset sample

Reset all

Simulation

Saved-run analysis

Run a configured 100% realization or explicitly evaluate the canonical unconfigured 150% model in an architecture context.

Preliminary engineering estimate — not a verified detailed-design result.

Prepare run

ANALYSIS MODE

Configured 100% model

Architecture-only 150% model

CONFIGURATION

Essential Capsule · valid

RUN NAME (EDITABLE BEFORE RUNNING)

Essential Capsule — 2026-08-17 09:03

Analysis basis: Essential Capsule · explicit realization available · 4 applied variation effect(s). A background realization is used whenever the explicit one is missing or stale.

KPI definitions

Formula dependencies must also be selected explicitly.

Total Mass

kg · kpi-mass

Estimated total cost

EUR · kpi-cost

Manufacturing Lead Time

h · kpi-lead-time

Resource Demand

resource-h · kpi-resource

Throughput Proxy

1/h · kpi-throughput

Total Power

kW · kpi-power

Additional direct algorithms

Use these only when no saved KPI definition is needed.

totalMass

directElementCost

processCost

estimatedTotalCost

totalPower

manufacturingLeadTime

Run preparation selects architecture, KPI basis and validation warnings.

Save Project

Load Project

150% Product-Line Model

Local autosave: Saved

Revision 1

Reset sample

Reset all

Input snapshot.

Premium Dual Boiler — comparable simulation

8/17/2026, 10:23:09 AM · rev 1

Current

Balanced Bean-to-Cup — comparable simulation

8/17/2026, 10:23:09 AM · rev 1

Current

Essential Capsule — comparable simulation

8/17/2026, 10:23:09 AM · rev 1

Current

Preliminary engineering estimate — not a verified detailed-design result. Source project revision 1.

Total Mass

KPI kpi-mass

650 kg

Formula / algorithm

totalMass

Input sources

Brewing and heating module / Automated mass: 650 kg (Robot-cell supplier budgetary data)

Assumptions

None

Missing information

None

Warnings

None

Estimated total cost

KPI kpi-cost

90016.45833333333 EUR

Formula / algorithm

estimatedTotalCost

Input sources

Brewing and heating module / Automated direct cost: 90000 EUR (Robot-cell supplier budgetary data); End-of-line test technician: 2 EUR/h (realized process duration 0.033333333333333 h (Test completed coffee machine) × rate 60 × quantity 1); Coffee-machine integration cell: Automated module assembly cell: 12.08333333333332 EUR/h (realized process duration 0.083333333333333 h (Integrate coffee-machine modules) × rate 145 × quantity 1); Module assembly workstation: Automated component feeder: 2.375 EUR/h (realized process duration 0.025 h (Manufacture coffee-machine modules) × rate 95 × quantity 1)

Assumptions

One engineering day is assumed to equal eight hours.

Missing information

None

Warnings

None

Manufacturing Lead Time

KPI kpi-lead-time

0.14166666666666666 h

The immutable result snapshot supports source-level review rather than presenting only final scalar values.

## Architecture + simulation acceptance checklist

| Control      | Acceptance evidence                                                                         |
|--------------|---------------------------------------------------------------------------------------------|
| Scope        | Intent and selected use cases satisfy the common completion predicates.                     |
| Architecture | All scoped functions have realizing product or industrial components.                       |
| Inputs       | Parameters, durations and resource quantities are complete, typed and sourced.              |
| Requirements | Bindings resolve and current evaluations show pass/fail/pending explicitly.                 |
| KPI basis    | Formula or algorithm, unit, direction, weight, objective and dependencies are explicit.     |
| Run          | Current immutable snapshot includes all selected numeric KPI results and reviewed warnings. |

# Chapter 3 - Trade study

Objective: derive at least two valid current 100% architectures from one canonical 150% family model, evaluate them on one requirement and KPI basis, and record an explicit approved decision.

## End-to-end workflow

| Step        | Authoritative activity                                                                             | Gate to next step                                            |
|-------------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| Problem 1-2 | Intent, SOL, scoped use cases and derived requirements.                                            | Common completion predicates satisfied.                      |
| Problem 3   | Decision question, active scope, variability axes, objectives, mandatory requirements and KPI set. | One shared evaluation basis.                                 |
| Solution 1  | Build full product and industrial 150% architecture.                                               | All common and variable functions/components trace-complete. |
| Solution 2  | Feature model and requires/excludes constraints.                                                   | Exactly one root and valid feature structure.                |
| Solution 3  | Feature expressions mapped to elements, relations or properties.                                   | Every variable architecture effect represented.              |
| Solution 4  | Create and validate configurations.                                                                | At least two meaningful valid, non-archived configurations.  |
| Solution 5  | Explicitly derive immutable 100% realizations.                                                     | At least two current realizations.                           |
| Solution 6  | Run same selected KPIs for every alternative.                                                      | Current complete simulation per realization.                 |
| Solution 7  | Feasibility, ranking, explicit selection and approval.                                             | Approved decision on a feasible alternative.                 |

The screenshot displays the MBSE / MBPLE Programme workbench interface. The sidebar on the left contains navigation options: Dashboard, PROBLEM SPACE (3/3), SOLUTION SPACE (7/7), and WORKSPACES (Model, Variability, Parameters and KPIs). The main workspace shows the 'ACTIVE PROJECT' 'Configurable Coffee Machine Product Line' with a dropdown menu and buttons for 'New project', 'Duplicate', 'Save Project', 'Load Project', and '150% Product-Line Model'. A status bar indicates 'Local autosave: Saved', 'Revision 1', 'Reset sample', and 'Reset all'. The 'PROJECT WORKFLOW' section for 'Configurable Coffee Machine Product Line' includes a warning: 'Preliminary engineering estimate — not a verified detailed-design result.' Below this, the 'Overall scope' section shows three options: 'Architecture building', 'Architecture building + simulation', and 'Trade study' (selected). The 'Status' bar indicates 'In progress' for the current step. The 'Problem Space' section shows the first three steps of the workflow: '1 Define mission, system and intent', '2 Define use-case and requirement scope', and '3 Define trade study, variability axes and evaluation KPIs'.

The Trade study dashboard sequence is authoritative. Do not jump directly from configuration to scoring.

## Study setup and shared evaluation basis

|   |                                                                                                                                                                                                            |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>Frame a resolvable decision</b><br><br>Enter study name and question. Select active use cases and the relevant needs/objectives. Mandatory requirements are collected through canonical derives traces. |
| 2 | <b>Define axes</b><br><br>Use independent architectural choices. The guided setup maintains one root feature and synchronizes each axis to a FeatureGroup.                                                 |
| 3 | <b>Fix KPI set</b><br><br>Select KPIs linked to the objectives. The same formulas, directions, units and weights apply to every alternative.                                                               |

Feasibility precedes utility

A requirement-infeasible alternative remains visible for learning but is ineligible to win. Weighted KPI scoring is applied only after feasibility; approval remains a human act.

The screenshot displays the 'Architecture Trade Study' interface. At the top, there's a navigation bar with 'Save Project', 'Load Project', and a dropdown menu set to '150% Product-Line Model'. Below this, a status bar indicates 'Local autosave: Saved', 'Revision 1', and buttons for 'Reset sample' and 'Reset all'. The main workspace is titled 'Architecture Trade Study' and includes a subtitle: 'Requirement feasibility first; weighted KPI comparison second; explicit user decision last.' The interface is divided into several sections: 'Trade Studies' with a '+ New' button and a list of active studies including 'Coffee-machine architecture selection' (3 alternatives, 1 result); 'Decision question' with fields for 'TRADE STUDY NAME' (Coffee-machine architecture selection) and 'DECISION QUESTION' (Which coffee-machine architecture should beco); and 'Problem-space scope' with fields for 'INHERITED SYSTEM OF INTEREST' (Coffee Machine System) and 'PROJECT WORKING SCOPE' (2 eligible use cases). A 'Run comparison' button is also visible.

Setup defines the decision boundary and the common requirement/KPI basis.

## 150% variability semantics

The canonical 150% architecture is not mutated during derivation. A configuration is a validated selection state; a 100% realization is an immutable audit artifact identifying included, excluded and modified model content at a specific source revision.

| Construct     | Purpose                                              | Key rule                            |
|---------------|------------------------------------------------------|-------------------------------------|
| Root feature  | Defines the family boundary.                         | Exactly one for the trade workflow. |
| Feature group | Encodes mandatory, optional or XOR choice structure. | Group cardinality must validate.    |

| Construct           | Purpose                                                               | Key rule                                                              |
|---------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------|
| requires / excludes | Encodes cross-tree dependencies and incompatibilities.                | Configurator may auto-apply requires; all conflicts must be resolved. |
| Variation point     | Maps a feature expression to an element, relation or property effect. | Preview common/included/excluded/modified status before derivation.   |
| Configuration       | Named 100% selection specification.                                   | Must be valid and non-archived.                                       |
| Realization         | Read-only derived 100% architecture and audit.                        | Records source revision; becomes stale after relevant change.         |

Save Project

Load Project

150% Product-Line Model

Local autosave: Saving

Revision 1

Reset sample

Reset all

## Variability

One canonical feature model drives validated selections and non-mutating 150% → 100% derivation.

Feature Model
Variation Points
Configurator
150% Preview
100% Realization
Derivation Summary

### Feature model

Containment is neutral; requires is green and directional; excludes is red. Organizational groups are recursive and nonselectable.

Graph

Table

+ Group

Constraint

+ Feature

#### Feature hierarchy

Containment defines the tree; cross-tree constraints are available on demand.

Automatic hierarc ▾

☐ Requires / excludes

Auto-arrange

The feature model constrains the legal family of 100% architectures.

**Variability**

One canonical feature model drives validated selections and non-mutating 150% → 100% derivation.

Feature Model **Variability Points** Configurator 150% Preview 100% Realization Derivation Summary

Graph List + Variation point

**Modeling sections**

All ☐ Mission and Context ☐ Requirements and Validation

☐ Product Functional ☐ Product Technical ☐ Process Functional

☐ Process Technical ☐ Interfaces and Traceability

**Element types**

All ☐ Mission ☐ Stakeholder ☐ Need ☐ Objective

☐ Use case ☐ System requirement ☐ Product function

☐ Product component ☐ Product interface ☐ Process function

41 element(s) and 74 relationship(s). A relationship is shown only when both endpoints are visible.

**Cross-domain variation-point graph**

Double-click an element body for Existence, a parameter or attribute for Property/Tag, or a connector for a relationship variation point. Existing mappings open for editing.

**Readable graph view**

Types occupy fixed top-down semantic bands.

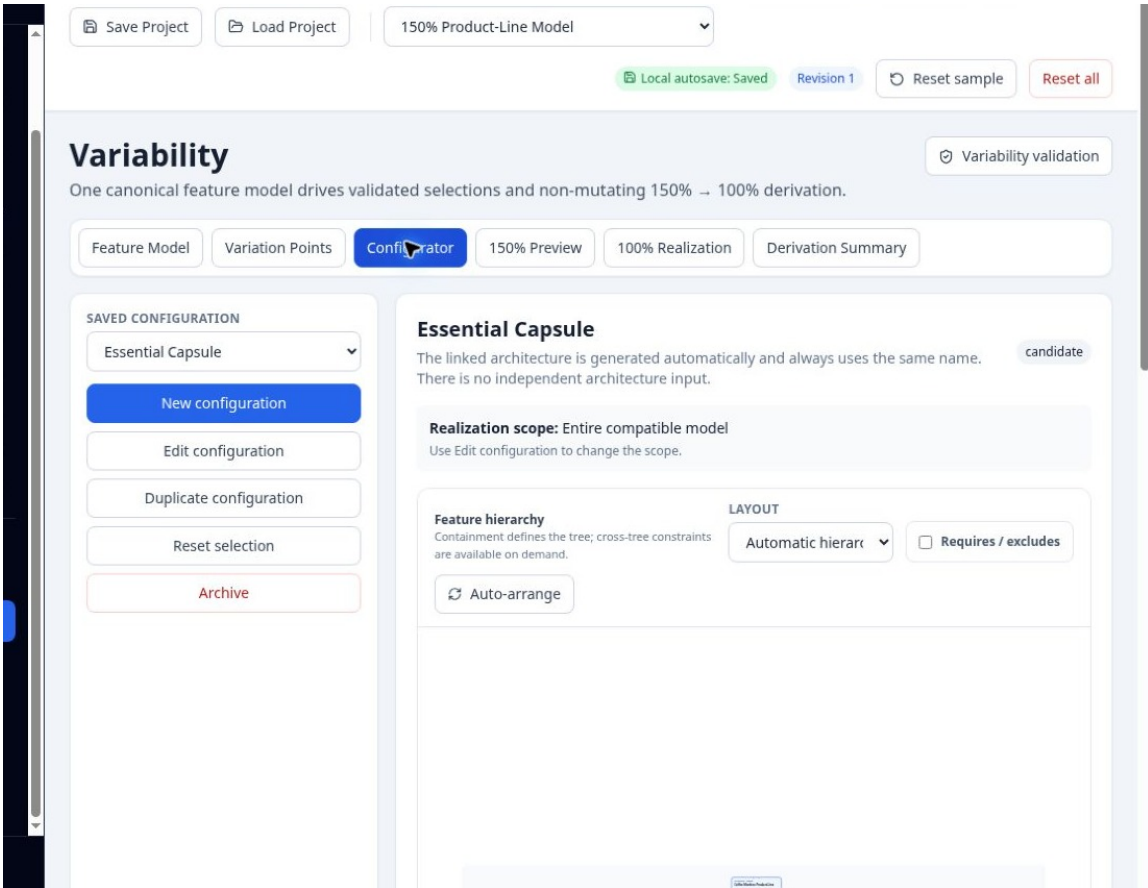
RELATIONSHIPS CARD DETAIL

Primary structure Compact Auto-arrange

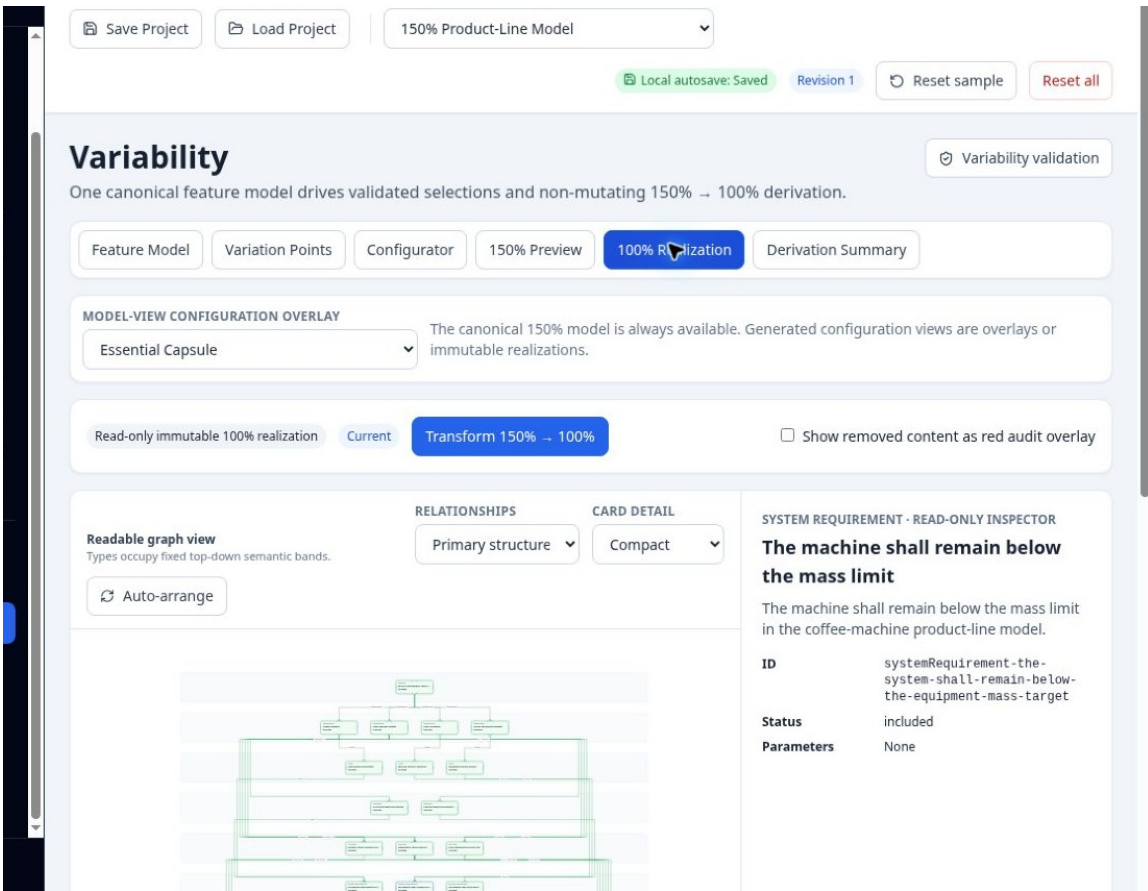
Mappings define how feature expressions suppress, include or modify architecture content.

## Configuration and derivation protocol

|   |                                                                                                                                                                                     |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>Validate feature selections</b> <p>Create at least two meaningful alternatives. Resolve group rules, requires and excludes until each is valid.</p>                              |
| 2 | <b>Derive explicitly</b> <p>Create a 100% realization for each valid configuration. Background in-memory realization during simulation is not sufficient for trade eligibility.</p> |
| 3 | <b>Audit structural consequences</b> <p>Review included/excluded/modified elements and relationships. Verify functions are not orphaned and required interfaces remain.</p>         |
| 4 | <b>Maintain current lineage</b> <p>After a 150% model, feature model or mapping change, revalidate and rederive before rerunning simulations.</p>                                   |



A valid configuration is a prerequisite, not a substitute, for an explicit 100% realization.



Realization audit shows the effective 100% content and its source revision.

### Configured simulation and eligibility

For each alternative select Configured 100% model and its current realization. Use the identical selected KPI set. The trade study considers only alternatives satisfying every eligibility gate below.

| Eligibility gate     | Required state                                                                      |
|----------------------|-------------------------------------------------------------------------------------|
| Configuration        | Valid and not archived.                                                             |
| Realization          | Explicit, current, generated from the applicable 150% revision.                     |
| Simulation           | Current, complete and attached to that realization/configuration.                   |
| KPI coverage         | Numeric result for every KPI selected by the study, including formula dependencies. |
| Requirement evidence | Mandatory requirement formulas resolved sufficiently to establish feasibility.      |

## Comparison and decision protocol

|   |                                                                                                                                                                                 |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>Create current comparison</b><br>Select eligible alternatives and compute the result. The result is immutable; later settings changes make it stale but retain it for audit. |
| 2 | <b>Review Manager view</b><br>Inspect feasibility, raw KPI values, normalized aggregate result and rank.                                                                        |
| 3 | <b>Review Expert view</b><br>Inspect per-KPI direction, normalization, global weight, contribution, formula/provenance and requirement evidence.                                |
| 4 | <b>Challenge sensitivity</b><br>Check whether small changes in weights or uncertain inputs change the ranking. Treat normalized scores as decision support, not truth.          |
| 5 | <b>Approve explicitly</b><br>Create decision from the current comparison, select a feasible alternative, record owner and rationale, confirm baseline approval and approve.     |

Save Project Load Project 150% Product-Line Model

Local autosave: Saved Revision 1 Reset sample Reset all

Coffee-machine architecture selection  
3 alternatives · 1 results

Duplicate

### Select configuration-derived alternatives

Step 7 begins here. Only valid configurations with current saved 100% derivations and complete simulations for every selected KPI are eligible.

- ☒ **Essential Capsule**  
Ready · current simulation Essential Capsule — comparable simulation
- ☒ **Premium Dual Boiler**  
Ready · current simulation Premium Dual Boiler — comparable simulation
- ☒ **Balanced Bean-to-Cup**  
Ready · current simulation Balanced Bean-to-Cup — comparable simulation

3/2 minimum alternatives selected

DECISION AT A GLANCE

**Which coffee-machine architecture should become the product-line baseline?**

Requirement feasibility is evaluated before KPI scoring. Infeasible alternatives remain visible but cannot be preferred.

| ALTERNATIVES | FEASIBLE | HIGHEST FEASIBLE SCORE      |
|--------------|----------|-----------------------------|
| 3            | 3        | <b>Balanced Bean-to-Cup</b> |

Manager view communicates feasibility and outcome without hiding infeasible alternatives.

Save ProjectLoad Project150% Product-Line Model

Local autosave: SavingRevision 1Reset sampleReset all

23T08:00:00.000Z

| Alternative          | KPI                     | Raw evidence | Normalized | Global weight | Contribution | Formula an                                                                         |
|----------------------|-------------------------|--------------|------------|---------------|--------------|------------------------------------------------------------------------------------|
| Essential Capsule    | Total Mass              | 420.00 kg    | 100.00     | 1             | 160.00       | 420 transform 80; raw weigh 4e73-a281-605 90cfa8f4-66a4 totalMass.                 |
| Essential Capsule    | Estimated total cost    | 35010.25 EUR | 100.00     | 1             | 268.39       | 35010.25 tran: function to 89. a3f8da54-edcc derivation-90c KPI formula es         |
| Essential Capsule    | Manufacturing Lead Time | 0.18 h       | 0.00       | 1             | 200.00       | 0.1833333333 minimize valu simulation-a3f 6053efd5106b 453c-85fa-44a manufacturing |
| Essential Capsule    | Throughput Proxy        | 5.45 1/h     | 0.00       | 1             | 43.64        | 5.4545454545 value function simulation-a3f 6053efd5106b 453c-85fa-44a throughputPr |
| Balanced Bean-to-Cup | Total Mass              | 420.00 kg    | 100.00     | 1             | 160.00       | 420 transform 80; raw weigh 47d5-bb67-22 8053a047-06e formula totall               |
| Balanced Bean-to-Cup | Estimated total cost    | 35016.46 EUR | 99.99      | 1             | 268.37       | 35016.458333 value function simulation-45 22ab379ccb91 06eb-4bbb-9b estimatedTota  |
| Balanced             | Manufacturing           | 0.14 h       | 100.00     | 1             | 200.00       | 0.1416666666                                                                       |

Expert view exposes the scoring mechanism and source evidence for review.

Save ProjectLoad Project150% Product-Line Model

Local autosave: SavedRevision 1Reset sampleReset all

TRACEABLE DECISION WORKFLOW

Manager viewExpert viewPreliminary engineering estimate — not a verified detailed-design result.

Architecture Trade Study

Requirement feasibility first; weighted KPI comparison second; explicit user decision last.

Trade Studies

+ New

ACTIVE TRADE STUDY

Coffee-machine architecture

Coffee-machine architecture selection

3 alternatives · 1 results

Duplicate

1. Define study2. Inspect evidence3. Decide

Run comparison

SAVED IMMUTABLE RESULT

7/23/2026, 10:00:00 AM · fixed-smart-mavt

Decision and rationale

approved

Only requirement-feasible alternatives are eligible. One feasible alternative is sufficient, with an explicit no-competitive-ranking notice.

SELECTED FEASIBLE ALTERNATIVE

Balanced Bean-to-Cup

OWNER

Sample Product Governance Board

RATIONALE

Sample governance decision: mandatory feasibility, Pareto position, fixed stakeholder value, sensitivity and bounded robustness were reviewed together.

Approval records deliberate selection, ownership and rationale; rank alone does not create a baseline.

## Trade-study acceptance checklist

| Control              | Acceptance evidence                                                                       |
|----------------------|-------------------------------------------------------------------------------------------|
| Decision boundary    | Question, scope, intent, mandatory requirements and KPI basis are explicit.               |
| 150% integrity       | Product and industrial family content is trace-complete before variation.                 |
| Feature validity     | One root, valid group cardinalities and resolved cross-tree constraints.                  |
| Mapping coverage     | Every variable architecture effect is mapped and previewed.                               |
| Alternative currency | At least two valid configurations, current realizations and current complete simulations. |
| Feasibility          | Requirement evidence is evaluated before score eligibility.                               |
| Scoring transparency | Raw values, normalization, weights, contribution and provenance are reviewable.           |
| Decision governance  | A feasible alternative is approved with owner and rationale from a current comparison.    |

# Technical reference - formulas, evidence and export

## Formula examples

| Purpose              | Example                                  | Notes                                                                 |
|----------------------|------------------------------------------|-----------------------------------------------------------------------|
| Calculated parameter | @length * @width * @height               | Local/refiner reference graph; use requested output unit if required. |
| Requirement          | @equipment_mass <= 1000                  | Exactly one comparator; stable binding behind symbol.                 |
| Direct KPI           | param("mass-a") + param("mass-b")        | Stable IDs; addition requires equal units.                            |
| Aggregated KPI       | sum(param("cost-a"), param("cost-b"))    | All arguments must have exactly equal units.                          |
| Dependent KPI        | kpi("direct-cost") + kpi("process-cost") | Both dependent KPIs must be selected in the run.                      |

Build the mission-to-validation thread in any order; the workflow and validation remain live.

System requirement + Add row + Add column Units

Mission and Context **Requirements and Validation** Product Functional Product Technical Process Functional Process Technical Interfaces and T...

Search elements, tags, parameters... All types All statuses

150% Product-Line Model All scopes Filter tag

Edit table Trace diagram Traceability matrix Model quality Manual order 5 of 41 elements

| Name                                          | Type                |
|-----------------------------------------------|---------------------|
| The machine shall meet the required beverage  | System requirement  |
| The machine shall verify beverage quality     | System requirement  |
| The machine shall remain below the mass limit | System requirement  |
| Manufacturing cycle-time analysis             | Verification method |
| End-of-line functional test                   | Verification method |

TAGS (COMMA-SEPARATED; EXISTING TAGS ARE SUGGESTED)

systemRequirement

OWNER SOURCE

Systems Engineering Coffee-machine demo

**Requirement formula**

Use arithmetic, parentheses and one comparison. References are stored by immutable ID. Manually entered variables may be saved, but remain pending and count as failed until assigned to a parameter or future KPI.

@equipment\_mass <= 1000

Select parameter or future KP + Insert

Requirement formula bindings connect numeric model evidence to pass/fail evaluation.

## Evidence interpretation

| Evidence                   | Review question                                                                                       |
|----------------------------|-------------------------------------------------------------------------------------------------------|
| Actual versus threshold    | Does the bound source value, unit and comparator match the intended verification rule?                |
| KPI source list            | Did the active model include only intended elements and applicable parameters?                        |
| Warning / assumption       | Is the approximation acceptable for the study authority and consistently applied across alternatives? |
| Breakdown / critical chain | Can the aggregate be traced to component, process and resource contributions?                         |
| Revision / stale marker    | Was the evidence generated after the latest relevant model and variability change?                    |

## Persistence and export

- Complete project JSON includes model, variability, configurations, realizations, simulation runs, trade studies and decisions. It excludes UI preferences and application snapshots.
- Selective JSON/XLSX export supports scoped review. Relationship closure automatically includes endpoints or records omissions when closure cannot be maintained.
- Built-in PDF reports are review outputs; the complete JSON remains the reloadable project artifact.
- Local snapshots are browser-local and limited. Use downloaded complete JSON for deliberate baselines and transfer.

## Common failure modes

| Failure                 | Diagnostic                                                                        | Recovery                                                               |
|-------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Requirement pending     | Missing binding, missing numeric value or unit/formula problem.                   | Resolve symbol binding and source parameter, then reevaluate.          |
| KPI formula invalid     | Missing stable ID, incompatible units or unselected KPI dependency.               | Reinsert references with builder, align units and select dependencies. |
| Process KPI undercounts | Process duration, realizedBy, requiresResource quantity or resource rate missing. | Complete the industrial evidence chain.                                |
| Lead time unexpected    | precedes network or duration units incorrect; parallel path misunderstood.        | Inspect sequence/graph and critical-chain breakdown.                   |
| Realization stale       | 150% model, feature model or mapping changed.                                     | Validate configuration and explicitly rederive.                        |
| Alternative ineligible  | At least one configuration, realization, simulation or KPI gate fails.            | Repair gates in that order; do not bypass eligibility.                 |
| Comparison stale        | Study basis or alternative evidence changed.                                      | Create a new current immutable comparison.                             |

### Expert review principle

Every decision-grade scalar should be navigable to active model content, stable formula references, units, assumptions, requirement evidence and the revision that produced it.

# Architect and Modeler answer handbook - expert edition

This branch-accurate handbook supplements the established guide. It explains what an unfamiliar user should enter, what the answer creates or changes, and when a question is review-only. Dynamic questions repeat for each item in the project.

## Handbook index

[How to answer Architect questions](#)

- [Study setup and intent \(AV-A02 to AV-B05\)](#)
- [Use cases and working scope \(AV-C01 to AV-C05\)](#)
- [Requirements \(AV-D01 to AV-D10\)](#)
- [Trade framing and early KPI intent \(Trade-off only: AV-T01 to AV-T06, then AV-I01 to AV-I07\)](#)
- [Product behavior and product architecture \(AV-E01 to AV-F06\)](#)
- [Industrial behavior and industrial architecture \(AV-G01 to AV-G12\)](#)
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- [Simulation questions \(Architecture and simulation: AV-I01 to AV-I12\)](#)
- [Variability and 100% configurations \(Trade-off only: AV-J01 to AV-K05\)](#)
- [Simulate, compare and decide \(Trade-off only: AV-S01 to AV-L10\)](#)
- [Final overview \(AV-M01 to AV-M06\)](#)

[Using Modeler view safely](#)

[Quick answer-quality checklist](#)

## How to answer Architect questions

The number of visible questions changes with the chosen scope and with the items you create. Questions marked "per item" repeat for each stakeholder, use case, requirement, function, component, KPI, feature axis or alternative.

### Study setup and intent (AV-A02 to AV-B05)

| Question                        | What to enter or select                                                                                                  | What it changes                                                                                    |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| AV-A02 - study scope            | Choose Architecture definition, Architecture and simulation, or Trade-off according to the decision you need to support. | Sets how far the guided workflow continues. Existing project data is preserved when scope changes. |
| AV-A03 - study name             | A short unique project name, such as Office coffee-machine concept.                                                      | Renames the active project.                                                                        |
| AV-A04 - mission                | One clear sentence describing the system's purpose and the problem it solves. Avoid listing a preferred solution.        | Creates or updates the Mission element.                                                            |
| AV-B01 - stakeholders           | Semicolon-separated people, organisations or external systems whose concerns matter. Include the system being designed.  | Creates Stakeholder elements and mission-to-stakeholder links.                                     |
| AV-A05 - system of interest     | Select exactly one stakeholder that represents the system being designed or analysed.                                    | Marks that stakeholder as the system of interest.                                                  |
| AV-B02 - stakeholder role       | Describe why each stakeholder matters.                                                                                   | Updates documentation only; it creates no relationship and changes no calculation.                 |
| AV-B03 - stakeholder needs      | Desired outcomes or problems from that stakeholder's viewpoint, separated by semicolons.                                 | Creates Need elements linked to the stakeholder.                                                   |
| AV-B04 - stakeholder objectives | Measurable improvements or targets, separated by semicolons.                                                             | Creates Objective elements linked to the stakeholder.                                              |
| AV-B05 - intent recap           | Confirm only after mission, stakeholders, needs and objectives describe the study correctly.                             | Records review; it does not override validation findings.                                          |

### Use cases and working scope (AV-C01 to AV-C05)

| Question                  | What to enter or select                                                                                         | What it changes                                    |
|---------------------------|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| AV-C01 - use cases        | Short action-oriented scenarios, separated by semicolons, such as Prepare beverage; Perform maintenance.        | Creates Use Case elements.                         |
| AV-C02 - participants     | For each use case, select all involved stakeholders, including the system of interest where applicable.         | Creates use-case participation links.              |
| AV-C03 - addressed intent | Select only the needs/objectives actually addressed by that use case.                                           | Creates canonical addresses relationships.         |
| AV-C04 - analysis scope   | Select the use cases this study will analyse. Only eligible cases involving the system of interest are offered. | Sets the persistent working use-case scope.        |
| AV-C05 - scope recap      | Check participants, addressed intent and active cases.                                                          | Records review and exposes missing scope evidence. |

## Requirements (AV-D01 to AV-D10)

| Question                                        | What to enter or select                                                                                                                                                | What it changes                                                                  |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| AV-D01/D02 - requirements from needs/objectives | Testable statements. Reuse an existing requirement when it expresses the same obligation; otherwise add a new one.                                                     | Creates System Requirement elements and canonical derives links.                 |
| AV-D03 - relevant use cases                     | Select the active use cases in which each requirement applies.                                                                                                         | Stores guided applicability context.                                             |
| AV-D04 - quantitative or qualitative            | Select quantitative when a number and comparison can test the requirement; qualitative when inspection, demonstration or another non-numeric judgement is appropriate. | Chooses the formula or verification path.                                        |
| AV-D05 - quantitative check                     | Enter measured property, semantic key, operator, target and unit. Example: mass, semantic key mass, <=, 12, kg.                                                        | Creates a formal requirement formula intent. It does not invent an actual value. |
| AV-D06 - actual-value source                    | Select the function/component that owns the measured property, or leave ownership for later architecture work.                                                         | Binds the formula intent to an authoritative model owner when one exists.        |
| AV-D07 - preliminary value                      | Enter a value only when supported; include unit, source, origin and optional uncertainty/range. Otherwise leave it pending.                                            | Creates or updates the owner's parameter and evidence fields.                    |
| AV-D08 - qualitative verification               | Name an analysis, inspection, demonstration or test and describe what will be checked.                                                                                 | Creates or updates a Verification Method.                                        |
| AV-D09 - importance                             | Mandatory means failure makes an alternative infeasible; Important and Desirable express lower decision priority.                                                      | Sets the requirement class.                                                      |
| AV-D10 - requirement recap                      | Confirm origins, wording, formula/verification path and ownership.                                                                                                     | Records review without forcing a passing result.                                 |

## Trade framing and early KPI intent (Trade-off only: AV-T01 to AV-T06, then AV-I01 to AV-I07)

| Question group                  | What to enter or select                                                                        | What it changes                                                       |
|---------------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| AV-T01/T02 - study and decision | A short Trade Study name and one explicit decision question comparing feasible concepts.       | Creates/updates the active Trade Study and its open decision context. |
| AV-T03 - common content         | Capabilities or solution elements required in every alternative; exclude things that vary.     | Proposes mandatory common features.                                   |
| AV-T04 - variation areas        | Independent choice areas such as brewing technology or interface type, not whole alternatives. | Proposes reusable variability axes.                                   |
| AV-T05 - concepts               | At least two alternative names with short descriptions.                                        | Creates configuration intents used later.                             |
| AV-T06 - framing recap          | Verify decision, common content, variation axes and concepts.                                  | Records review.                                                       |
| AV-I01 - evaluation results     | Select only KPIs needed to judge the objectives.                                               | Creates/selects canonical KPI definitions; it does not run analysis.  |
| AV-I02 - KPI purpose            | Link each KPI to the needs/objectives it measures.                                             | Updates KPI objective links.                                          |
| AV-I03 - calculation method     | Use a standard algorithm when its meaning and inputs fit; otherwise choose a guided formula.   | Sets the KPI calculation mode.                                        |
| AV-I06 - interpretation         | Enter output unit, minimize/maximize direction, target and optional acceptable bounds.         | Defines how the KPI is read.                                          |
| AV-I07 - global weight          | Enter a finite non-negative weight. At least one selected KPI must be greater than zero.       | Sets the reusable KPI default weight.                                 |

## Product behavior and product architecture (AV-E01 to AV-F06)

| Question group                           | What to enter or select                                                                                                                  | What it changes                                                                                                     |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| AV-E01/E02 - product functions and order | For each use case, enter product functions in normal execution order; then correct the proposed sequence if needed.                      | Creates Product Function elements, use-case hasFunction links, a contextual sequence and cycle-free precedes links. |
| AV-E03 - function satisfaction           | Select requirements directly helped by each function; leaving a function unlinked is allowed when justified.                             | Creates requirement satisfiedBy links.                                                                              |
| AV-E04 - product-behavior recap          | Check use-case coverage, sequence and requirement links.                                                                                 | Records review.                                                                                                     |
| AV-F01 - realizing components            | Name one or more physical/logical product components that perform each function.                                                         | Creates Product Component elements and realizedBy links.                                                            |
| AV-F02 - component satisfaction          | Select requirements directly satisfied by each component.                                                                                | Creates requirement satisfiedBy links.                                                                              |
| AV-F03/F04 - parameter owner/value       | Select the authoritative product function/component for a pending measured property; then enter value, unit, source and origin if known. | Creates or reuses one canonical parameter and binds the requirement formula.                                        |
| AV-F05 - product interaction             | When useful, select another product component, name the interface and describe exchanged material, energy or information.                | Creates Interface content and canonical links.                                                                      |
| AV-F06 - product-architecture recap      | Check function realization, interfaces, parameters and satisfaction.                                                                     | Records review and exposes gaps.                                                                                    |

## Industrial behavior and industrial architecture (AV-G01 to AV-G12)

| Question group                           | What to enter or select                                                                                                                                                            | What it changes                                                                                       |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| AV-G01/G02 - process functions and order | For each use case, enter industrial-system process functions such as positioning, fastening or inspection, then verify their order. Leave blank for a genuinely product-only case. | Creates Process Function elements, hasFunction, sequences and precedes links.                         |
| AV-G03 - process satisfaction            | Select requirements each process function helps satisfy.                                                                                                                           | Creates requirement satisfiedBy links.                                                                |
| AV-G04 - realizing equipment             | Name the workstation, equipment or plant element that performs each process function.                                                                                              | Creates Industrial System Components and realizedBy links.                                            |
| AV-G05/G05A/G05B - industrial evidence   | Select satisfied requirements; resolve any property owned by an industrial function/component and enter evidence if known.                                                         | Creates satisfaction links and canonical parameters.                                                  |
| AV-G06 - duration                        | Enter the authoritative process duration, time unit, source, origin and optional uncertainty.                                                                                      | Updates process metadata used by lead-time, cost and resource calculations.                           |
| AV-G07/G08 - consumed/produced product   | Select product components and enter positive quantity and unit.                                                                                                                    | Creates canonical consumes and produces flows.                                                        |
| AV-G09/G10 - resources                   | Name required people, roles, skills, tools, machines, software or facilities; then enter quantity, unit, optional rate, capacity and availability.                                 | Creates Resource elements and industrial-component requiresResource links with quantitative metadata. |
| AV-G11 - industrial interaction          | Add only meaningful material, energy or information exchanges between industrial components.                                                                                       | Creates interface content and links.                                                                  |
| AV-G12 - industrial recap                | Check realization, sequence, handoffs, duration and resources.                                                                                                                     | Records review and exposes model-quality gaps.                                                        |

## Traceability and readiness (AV-H01 to AV-H04)

| Question                        | What to enter or select                                                                                                          | What it changes                                                                                      |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| AV-H01 - verification           | For every requirement, identify a credible analysis, inspection, demonstration or test; optionally allocate a process function.  | Creates/updates Verification Methods and verifies/allocation links.                                  |
| AV-H02 - satisfying evidence    | Select at least one compatible function, technical component or existing owned parameter. Do not create a second parameter here. | Creates satisfaction links or binds the selected parameter and its owner to the requirement formula. |
| AV-H03 - evidence credibility   | Confirm only when value, target, unit, source, origin and verification are credible; otherwise mark pending.                     | Records an explicit evidence-review state.                                                           |
| AV-H04 - architecture readiness | Review intent, behavior, realization, satisfaction, flows, resources and verification.                                           | Calculates readiness from the model; confirmation cannot hide blockers.                              |

## Simulation questions (Architecture and simulation: AV-I01 to AV-I12)

After choosing KPIs and their purpose/calculation/interpretation as described above:

| Question group                 | What to enter or select                                                                                                | What it changes                                            |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| AV-I04/I05 - exact calculation | Confirm the inputs used by a standard algorithm, or build a formula by inserting named parameters/KPIs.                | Stores no duplicate values; formulas keep immutable IDs.   |
| AV-I08 - missing inputs        | Select every authoritative contributing owner and enter separate value, unit, source, origin and optional uncertainty. | Creates/updates canonical parameters used by the KPI.      |
| AV-I09 - context               | Select the current canonical architecture. Existing current 100% configurations may also appear when available.        | Chooses the exact model context for the run.               |
| AV-I10 - assumptions           | Read and acknowledge assumptions/non-blocking warnings when presented.                                                 | Stores acknowledgement with run evidence.                  |
| AV-I11 - run                   | Enter a meaningful run name and execute.                                                                               | Validates inputs and creates one immutable Simulation Run. |
| AV-I12 - recap                 | Confirm only after KPI results, requirement status, warnings and sources match the intended analysis.                  | Records review of current evidence.                        |

## Variability and 100% configurations (Trade-off only: AV-J01 to AV-K05)

| Question group                             | What to enter or select                                                                                                                               | What it changes                                                               |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| AV-J01/J02 - family and common features    | Name the complete product line and confirm mandatory capabilities shared by all alternatives.                                                         | Creates one Root Feature and mandatory child features.                        |
| AV-J03/J04 - axes and choices              | For each variation area, list choices and select XOR, OR, optional or typed mode; list allowed values for typed axes.                                 | Creates synchronized Variability Axes, FeatureGroups and Features.            |
| AV-J05 - constraints                       | Add only necessary requires or excludes rules.                                                                                                        | Prevents invalid feature selections.                                          |
| AV-J06 - element applicability             | Map only genuinely variable requirements/functions/components/processes/resources/verification methods to a feature; unlisted elements remain common. | Creates existence applicability using the existing variation mechanism.       |
| AV-J07/J08 - property variation and domain | Map a feature to a changed existing parameter value and select affected model domains when needed.                                                    | Creates typed Variation Points with exact target and realization scope.       |
| AV-J09 - 150% recap                        | Check family completeness, hierarchy, constraints, applicability and property changes.                                                                | Records review.                                                               |
| AV-K01 - configuration                     | For each concept, choose one value for every required group and any optional features.                                                                | Creates/updates a Configuration.                                              |
| AV-K02/K03 - invalid selection             | Apply a proposed required-feature correction or return to the choices.                                                                                | Keeps invalid drafts visible but blocks derivation/simulation until valid.    |
| AV-K04 - derive                            | Review included, excluded and modified content, then derive.                                                                                          | Creates immutable 100% realization evidence without changing the 150% source. |
| AV-K05 - realization recap                 | Compare selected features with included/excluded elements and applied property changes.                                                               | Confirms the current derivation.                                              |

## Simulate, compare and decide (Trade-off only: AV-S01 to AV-L10)

| Question group                         | What to enter or select                                                                                                           | What it changes                                                                                         |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| AV-S01 - readiness                     | Verify every intended alternative has a valid configuration and current 100% derivation.                                          | Identifies blockers; it creates no run.                                                                 |
| AV-S02/S03/S04 - comparable simulation | Acknowledge shared assumptions, run the same KPI set for every alternative, then review coverage, units and sources side by side. | Atomically creates one immutable configured Simulation Run per alternative; no partial batch is stored. |
| AV-L01/L02/L03 - comparison set        | Select at least two eligible alternatives, mandatory feasibility requirements and comparable KPIs.                                | Sets the active study's evidence basis.                                                                 |
| AV-L04/L05 - priorities and limits     | Set study-specific KPI weights, preferred direction and optional warning/hard thresholds.                                         | Updates study settings without changing global KPI definitions.                                         |
| AV-L06/L07/L08 - compare               | Resolve blocking comparability gaps, run the calculation and review feasibility, weighted scores, ties and sensitivity.           | Stores an immutable Comparison Result. Infeasible alternatives remain visible but cannot be preferred.  |
| AV-L09/L10 - decision                  | Explicitly select a feasible baseline, record rationale, owner, status/date and confirm approval.                                 | Creates/updates the Decision and changes the baseline only after explicit approval.                     |

## Final overview (AV-M01 to AV-M06)

The final pages are reviews, not new data-entry steps. They summarise the study question, model content, stakeholder traceability, requirements, architecture, current simulation and - for Trade-off - alternatives and decision. The completion banner is green only when required answers and evidence are complete and reviewed; amber means pending evidence, warnings, skipped/incomplete answers or answers needing review; red means a failed requirement or another blocking model finding.

## Using Modeler view safely

Use the Dashboard to find the next activity. The Model workspace edits canonical elements and relationships through Table, Graph, Sequence, Matrix and Model Quality views. Variability edits the feature model, variation points, configurations and 150%/100% realizations. Parameters and KPIs edits authoritative engineering inputs and formulas. Simulation creates immutable runs. Architecture Trade Study frames, compares and decides. Delivery saves projects, snapshots and exports. Model Quality collects project-wide findings and links back to affected content.

When an Architect question is unclear, switch to Modeler view to inspect the relevant table, matrix or graph, then return. If a Modeler edit changes content used by a confirmed answer, the affected question and recap are marked for review; unrelated documentation edits stay confirmed. Never duplicate an existing parameter merely to satisfy a question: identify its authoritative owner and reuse the canonical value.

## Quick answer-quality checklist

- Use stakeholder language for needs and measurable language for objectives.
- Write requirements so a verification method or formula can decide them.
- Give every numeric value a unit, source and value origin.
- Keep product functions distinct from the product components that realize them.
- Keep industrial process functions distinct from workstations/equipment that realize them.
- Allocate resources to the industrial component that requires them.
- Treat optional content and changed values as variability; keep common content unconditioned.
- Compare only current 100% derivations with current, complete and comparable simulation evidence.
- Record a decision rationale; do not treat the highest score as automatic approval.